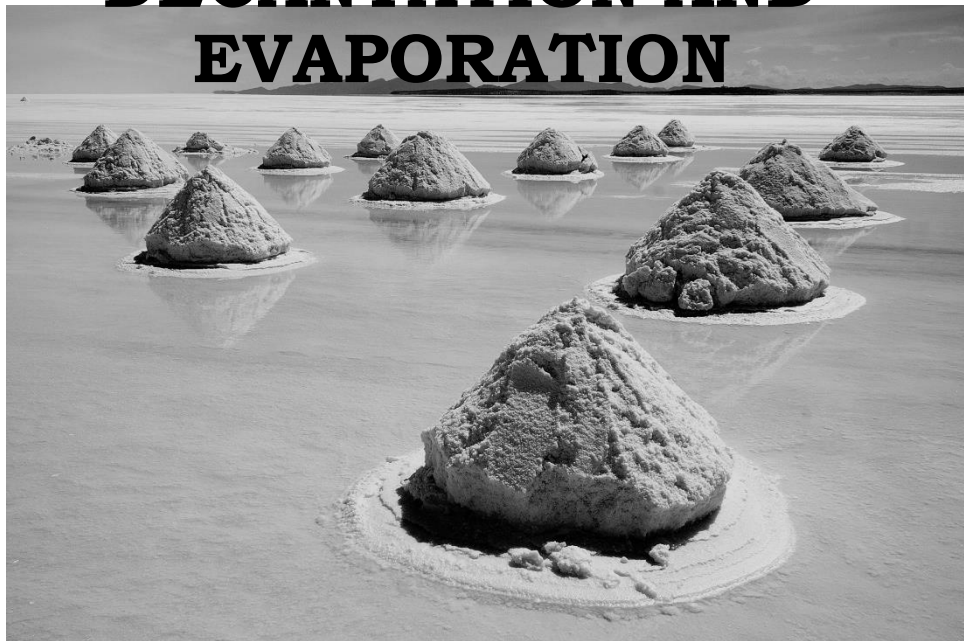


Science

Quarter 1– Module 4

SEPARATING MIXTURES: DECANTATION AND EVAPORATION



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Enumerate techniques in separating mixtures such as decantation, evaporation, filtering, sieving and using magnet
(S6MT-Id-f-2)

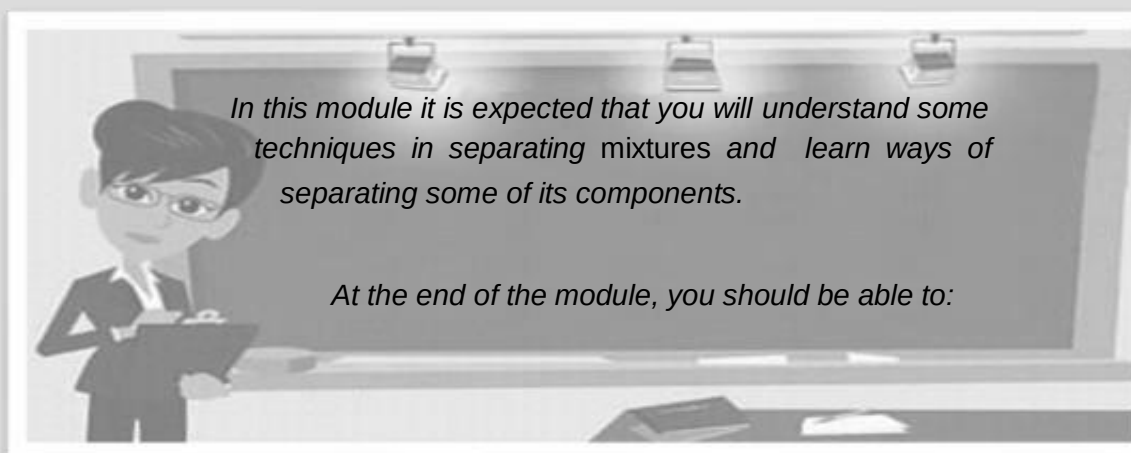


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SEPARATING MIXTURES: DECANTATION AND EVAPORATION



In the previous modules you have learned about the different types of mixtures that we encounter every day. You have also learned their properties and uses. Mixtures are very useful. However, sometimes we need to separate mixtures into their components. Why there is a need to do so? What techniques are we going to use in separating its components? All of these questions will be answered as we go through this module.



- describe decantation as a process of separating mixtures
- describe evaporation as a process of separating mixtures
- cite instances where decantation and evaporation are used as processes in separating mixtures and produce products which are beneficial to the community



Directions: Read and analyze each statement carefully. Write the letter of your answer in your notebook.

1. How can you remove floating oil in water?
a. by decantation b. by evaporation c. by filtration
2. Which of the following is the advantage of filtration from decantation?
I. Filtration is used for mixture of insoluble solid in liquid.
II. Filtration totally separates the solid material from liquid.
III. Filtration is easy to perform than decantation.
a. I only b. I and II only c. II and III only
3. After deep-frying a chicken, which method of separating mixtures is applicable to separate the small pieces of chicken meat from the oil?
a. decantation b. evaporation c. sieving

4. Which of the following is a practical application of evaporation?
- Film deposition
 - Evaporation of sweat from our body.
 - Drying of clothes by hanging them out in an open area.
- a. I only b. II only c. II and III only
5. Mixtures whose solid components settle down in a liquid can be separated by _____.
- a. evaporation b. decantation c. filtration

Did you find it difficult to answer the questions? Don't worry as you go on with this module you will learn more about this lesson. To check your answers you may refer to the answer key found on the last page of this module.



RIDDLE TIME!

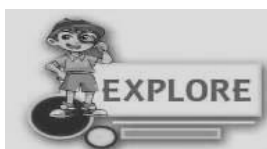
Directions: Read the riddle inside the picture. . With the help of the picture you can guess the correct jumbled words inside the box.



R	A	T	I	O	N	P	O	V	A	E
---	---	---	---	---	---	---	---	---	---	---



Did you find this interesting? If your answer in the first picture is EVAPORATION and DECANTATION in the second picture, you got this riddle correct. You are ready for the next activity.



DOES RICE SETTLE OR FLOAT?

Lesson Concept:

Decantation is the “pouring off” of a liquid from a solid/liquid mixture. The mixture is allowed to settle, and the liquid is removed while preventing the solid from escaping.

Materials

¼ kilo of rice
 ½ liter of Water
 transparent pitcher
 transparent bowl

Instructions:

- a. Gather all the materials needed.
- b. Put the rice in a bowl then add $\frac{1}{2}$ liter of water.
- c. Wash the rice with your hands. Leave the water to settle for 1 minute. What did you observe?
- d. Slightly tilt the bowl to pour out some water
- e. Observe then answer the following question in your activity notebook:

1. Describe your mixture?

2. How did you separate the impurities from the rice before cooking?

3. What happened to the rice? Why?

4. What happened to the impurities (husk etc.)? Why?

5. Can the solid particles be separated from the water without letting it settle at the bottom of the bowl? Why?

HEAVY AND LIGHT LIQUIDS!

LESSON CONCEPT:

Decantation can also be used for liquid mixtures. It is used when separating two or more immiscible liquids. Once the mixture components have separated by forming a layer between them in a container, the lighter (less dense) liquid is poured off leaving the heavier (denser) liquid behind.

MATERIALS:

Clear glass or cup container	Cooking oil	Kerosene
Soy sauce	Vinegar	Water

PROCEDURE:

1. Gather the materials needed.
2. Choose two liquids and mix them in an empty glass or cup container.
3. Leave the mixture for two to three minutes. Observe if the two liquids completely mixed with each other. If they did, they are called miscible liquids. Liquids, which do not mix together and form a layer between them are called immiscible liquids. Record your observations.

Take precautionary steps in handling kerosene. Ensure that its container is properly labeled and sealed. Wash your hands after holding it.

4. Describe the mixture formed. Which liquid submerged at the bottom of the container?
5. For immiscible liquids, try to separate it through decantation or by pouring or scooping into another container.

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Table 1. Miscible and Immiscible Liquids

Liquids		Miscible	Immiscible	Description	Liquid at the bottom of the Glass
1	2				

Guide Questions:

1. What mixture of liquids are immiscible? Miscible?

2. Did a longer waiting time improve the clarity of liquid mixtures? Why?

3. Explain how you were able to remove and separate the two immiscible liquids?

4. What other methods can be done to separate immiscible liquids?

EVAPORATE: LET'S EVAPORATE!

Reminder:

1. Always ask assistance from your parents/guardian. Don't do the activity alone.
2. Answer the guide questions in your activity notebook.

Objective:

At the end of the activity, learners should be able to explain when to use evaporation in separating mixtures.

Materials:

- Pinch of salt
- Alcohol lamp/hot plate or gas burner
- water
- spoon
- pan

Caution: Make sure to avoid contact with the hot plate/gas burner or the pan when it is hot to avoid getting burned.

Procedure:

1. Put 1 spoon of salt in a pan with 1 glass of water.
2. Stir the solution and cover once done.
3. Place pan at the gas burner and let it boil until no water is left in the pan.

(**Note:** If using hot plate, put the pan on top of the hot plate and set the temperature to hot mode.)



Guide Questions:

1. What happens to salt solution after the water evaporated ?
2. What was left in the pan?
3. In what other ways can we use evaporation in separating mixtures?

EVAPORATE!

Directions: Examine the pictures below and answer the guide questions that follow. Record your answer in your activity notebook.





Guide Questions:

1. What is the use of evaporation in the preparation of these products?
2. Cite some other examples where evaporation is useful in your community?



Decantation is a process of separating mixture by removing a liquid layer that is free from a precipitate. The purpose may be to obtain a decant (liquid free from particulates) or to recover the precipitate. Decantation relies on gravity to pull the precipitate out of a solution so there is always some loss of material or product, either from precipitate not fully falling out of the solution or from leaving some liquid when separating from the solid portion.

There are two main decantation methods:

1. Separating Liquids and Solids

Decanting is done to separate particulates from a liquid by allowing the solids to settle to the bottom of the mixture and pouring off the particle-free part of the liquid.

For example, a mixture (possibly from a precipitate reaction) is allowed to stand so that gravity has time to pull the solid to the bottom of a container. The process is called sedimentation. Using gravity only works when the solid is less dense than the liquid. Clearwater may be obtained from mud by simply allowing time for the solids to separate from the water.

The separation may be enhanced using centrifugation. If a centrifuge is used, the solid may be compacted into a pellet, making it possible to pour off the decant with minimal loss of liquid or solid.

1. Separating Two or More Liquids

Another method is to allow two immiscible liquids to separate and the lighter liquid is poured or siphoned off. A common example is decantation of oil and vinegar. When a mixture of the two liquids is allowed to settle, the oil will float on top of the water so the two components may be separated. Kerosene and water can also be separating using decantation.

The two forms of decantation may be combined. This is especially useful if it's important to minimize loss of a solid precipitate. In this case, the original mixture may be allowed to settle or may be centrifuged to separate the decant and sediment. Rather than immediately drawing off the liquid, a second immiscible liquid may be added that is denser than the decant and that doesn't react with the sediment. When this mixture is allowed to settle, the decant will float on top of the other liquid and sediment. All of the decant can be removed with minimal loss of precipitate (except a tiny amount that remains floating in the mixture). In an ideal situation, the immiscible liquid that was added has a high enough vapor pressure that it evaporates, leaving all of the sediment.

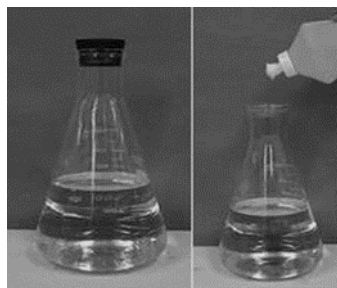
In everyday work, the term 'decantation' is usually associated with wine. Decanting is also a chemical laboratory process used to separate mixtures. Decanting is a process to separate mixtures.

Decanting is just allowing a mixture of solid and liquid or two immiscible liquids to settle and separate by gravity. This process can be slow and tiresome without the aid of a centrifuge. Once the mixture components have separated, the lighter liquid is poured off leaving the heavier liquid or solid behind. Typically, a small amount of the lighter liquid is left behind.

Some Mixtures That Can Be Decanted



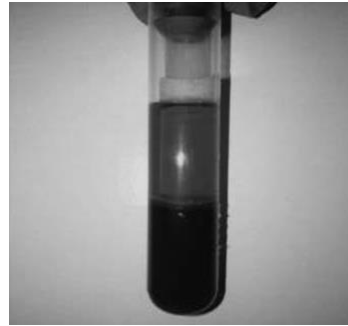
Oil and water – Oil floats on top of water. Decanting the mixture allows the oil to be poured off the water.



Gasoline or kerosene and water – this mixture is an example often cited as a safety hazard. Decanting a mixture containing flammable solvents can be dangerous as the flammable material evaporates and forms dangerous fumes.



Cream and milk – Cream is separated from milk by decantation. Cream rises to the top of the milk mixture and is easily skimmed off.



Blood and plasma – A centrifuge is necessary for this decantation. Plasma can be removed from blood by decantation



Dirt and water – muddy water can be cleared up by decanting. The soil will sink to the bottom of the tube allowing the clear water to be poured off.



Wine – Sediment from the fermentation process can produce an undesirable taste. Wine is decanted to separate the wine from these sediments



TEST A

Directions: Use the words in the box in completing the paragraph below.

mixture less-dense

scooping components

pouring denser technique undisturbed

Decantation is a 1. _____ used in separating a 2. _____ substance from a 3. _____ one. In the process of decantation, the 4. _____ is left 5. _____. When the 6. _____ are separated, the less-dense substance could be removed by 7. _____ using a spoon or it can be removed by slowly 8. _____ out the less-dense substance.

Test B

Directions: Read each item carefully. Choose the letter of the correct answer. Write your answer in your notebook.

- ___ 1. Salt is obtained from _____.
a. freshwater b. seawater c. rainwater
- ___ 2. What is an important factor needed in separating mixtures through evaporation?
a. light b. heat c. wind
- ___ 3. What will happen to the seawater if it is heated continuously?
a. turns into vapor b. turns into salt c. turns into water droplets
- ___ 4. Salt is dissolved in water. How can you separate the salt from the mixture?
a. filter the water b. dispose the water c. boil the water and let it evaporate
- ___ 5. The following mixtures can be separated through evaporation EXCEPT one.
a. mixture of salt and water
b. mixture of sand and water
c. mixture of sugar and water

Congratulations, you have finished this fourth module. Verify your answer in the answer key section of this module. If you still need to understand kindly go back to the first section. If you think you are ready to learn new topic proceed to the succeeding module after accomplishing the assessment found at the next section.



To show understanding of the concepts you learned in this module, write a rap or a poem about the process in separating liquid and solid mixtures through decantation and evaporation. Write it in a clean sheet of paper and submit it to your teacher.

Directions: Read carefully and understand the questions. Write **ONLY** the **LETTER** of your choice.

1. Salt making is based on the idea of _____.
a. crystallization c. evaporation
b. decantation d. filtration
2. How can you remove floating oil in water?
a. by decantation c. by evaporation
b. by filtration d. by sieving

3. Which of the following techniques is used to separate mixtures with nonvolatile solute dissolved in a solution?
- a. decantation
 - b. evaporation
 - c. filtration
 - d. sieving
4. Which statement is correct about evaporation as a process in fish drying?
- a. evaporation adds heat to the fish.
 - b. evaporation removes the salt in drying fish.
 - c. evaporation separates the liquid molecules from the fish.
 - d. evaporation had no connection to fish drying.
5. What is the main role of evaporation as a process in developing a product?
- a. evaporation as a process separates the liquid components in the mixture.
 - b. evaporation as a process absorbs water in the mixture.
 - c. evaporation as a process adds heat in the mixture.
 - d. evaporation as a process separates all solid components in the mixture
6. The following are useful processes of evaporation EXCEPT
- a. copra drying
 - b. fish drying
 - c. salt extraction
 - d. separation of color
7. Mixtures whose solid components settle down in a liquid can be separated by:
- a. evaporation
 - b. decantation
 - c. filtration
 - d. sieving
8. Salt is obtained from_____.
- a. freshwater
 - b. rain water
 - c. seawater
 - d. tap water

ANSWER KEY

TRY THIS

1. a
2. b
3. a
4. c
5. b

APPLY WHAT YOU HAVE LEARNED

Test A

1. technique
2. less-dense
3. denser
4. mixture
5. undisturbed
6. components
7. scooping
8. pouring

Test B

1. b
2. b
3. b
4. c
5. b

ASSESS WHAT YOU HAVE LEARNED

- | | |
|------|--------|
| 1. c | 6. d |
| 2. a | 7. b/c |
| 3. b | 8. c |
| 4. c | |
| 5. a | |

REFERENCES

Science Spectrum 4 by Fallaria et.al p. 156

The New Science Links 6 by Evelyn Castante-Padpad et.al

(2017 Edition) pp.87 and 91

Science in Our World 6 by Norma Garcia, et al 2014 p.24-25,40

LRMDS: Science 6 Quarter 1 week 4

Daily Lesson Log – Decantation

<https://www.scribd.com/doc/147308397/Lab-Decantation>

<https://foodrecap.net/health/cold-water-fat-float/>

<https://www.youtube.com/watch?v=PMB879NxGHU>

<https://www.thoughtco.com/definition-of-decantation-604990>

Grade 6 Science Activities First Quarter

(Week 1-10)

Activity 6.1 Dry It Up!

Helmenstine, A.M. Ph.D. January 05, 2019. How to Separate Salt and Sand - 3

Methods. Retrieved from <https://www.thoughtco.com/separating-salt-and-sand-4055888>

Prototyped & Contextualized DLP Grade 6 Science



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